

Equality cases in the sharp one-dimensional matrix Lieb–Thirring inequality

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Abstract

We classify all equality cases in the sharp one-dimensional Lieb–Thirring inequality for every finite matrix size and $1/2 < \gamma < 3/2$. For measurable Hermitian positive semidefinite potentials W with $\int_{\mathbb{R}} \text{tr}(W^{\gamma+1/2}) < \infty$, equality holds precisely for direct sums, in one constant unitary basis, of scalar one-bound-state solitons and zero channels. The nonzero solitons may have independent scales and centers.

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1 Introduction

Lieb–Thirring inequalities compare the binding energy of a Schrödinger operator with an integral of its potential. Equality asks how the potential must arrange that binding. In one dimension, a scalar one-bound-state optimizer has a hyperbolic-secant profile. For a matrix potential, several such profiles can occupy orthogonal internal channels. The issue is whether equality also permits channels that rotate with position, or several bound states that interact within the same channel. We prove that throughout $1/2 < \gamma < 3/2$ every equality case consists of independent scalar profiles in one fixed basis.

Let $m \geq 1$ be finite, let $1/2 < \gamma < 3/2$, and set $p = \gamma + 1/2$. For a measurable Hermitian positive semidefinite function $W : \mathbb{R} \rightarrow \mathbb{C}^{m \times m}$ satisfying

$$\int_{\mathbb{R}} \operatorname{tr}(W(x)^p) dx < \infty, \quad (1.1)$$

let $H_W = -d^2/dx^2 - W$ be the operator associated with

$$h_W[\psi] = \int_{\mathbb{R}} \|\psi'\|^2 - \int_{\mathbb{R}} \langle \psi, W\psi \rangle, \quad \psi \in H^1(\mathbb{R}; \mathbb{C}^m). \quad (1.2)$$

Lemma 2.1 proves that the form is closed and bounded below and that the negative spectrum is discrete, with possible accumulation only at zero. We write

$$\operatorname{Tr}(H_W)_-^\gamma = \sum_{\lambda_j(H_W) < 0} |\lambda_j(H_W)|^\gamma,$$

counting multiplicities. The finite-dimensional trace tr is unnormalized; matrix powers are defined by spectral functional calculus.

Theorem 1.1 (Equality classification). *For every W satisfying (1.1),*

$$\operatorname{Tr}(H_W)_-^\gamma \leq C_\gamma \int_{\mathbb{R}} \operatorname{tr}(W^p), \quad C_\gamma = \left(\frac{\gamma - 1/2}{\gamma + 1/2} \right)^{\gamma-1/2} \frac{\Gamma(\gamma + 1)}{\sqrt{\pi} \Gamma(\gamma + 3/2)}. \quad (1.3)$$

Put $r = (\gamma - 1/2)^{-1}$. Equality holds if and only if there are $k \in \{0, \dots, m\}$, a constant unitary matrix U , positive numbers $a_1, \dots, a_k$, and real numbers $x_1, \dots, x_k$ such that almost everywhere

$$W(x) = U \operatorname{diag}(w_1(x), \dots, w_k(x), 0, \dots, 0) U^*, \quad w_j(x) = (r + 1) a_j^2 \operatorname{sech}^2(r a_j (x - x_j)). \quad (1.4)$$

The case $k = 0$ means $W = 0$. Each nonzero channel has exactly one negative eigenvalue, $-a_j^2$, and therefore every extremal potential has at most m negative eigenvalues, counted with multiplicity.

The inequality in (1.3) is the sharp matrix inequality of [11, Theorem 1.1]. Our contribution is its equality classification. We retain the analytic constructions in the proof, so that the inequality and its equality conditions are established here under the same measurable potential hypothesis. The scales and centers in (1.4) are independent; repeated scales and identical profiles are allowed.

1.1 History and related results

Lieb and Thirring introduced spectral moment inequalities in their work on the kinetic energy of fermions and stability of matter [8, 9]. Their sharp constants distinguish the regimes in which binding is best concentrated in a single state or distributed among many states. Keller’s

optimization of the lowest Schrödinger eigenvalue [6] supplies the one-state problem underlying the scalar profiles in (1.4); their explicit form appears in [9, Section 4(B), Equations (4.19)–(4.20)]. The present question concerns equality in the sum of all negative eigenvalue moments, with a matrix potential that need not commute with itself at different points.

The sharp scalar endpoint $\gamma = 1/2$ was proved by Hundertmark, Lieb and Thomas [5]; Hundertmark, Laptev and Weidl [4, Theorem 3.1] extended it to operator-valued potentials. The semiclassical constant is sharp for $\gamma \geq 3/2$; the lifting of moments of Aizenman and Lieb [1] and the operator-valued inequalities of Laptev and Weidl [7] are central to this development. Benguria and Loss [2] gave a commutation proof of the matrix result at $\gamma = 3/2$. More recently, Read and Schulz [12] proved the sharp scalar inequality at $\gamma = 1$, with constant $4/(3\sqrt{3}\pi)$, and its operator-valued extension. The matrix companion [11] proves the sharp one-state constant for the whole open interval considered here.

The endpoint $\gamma = 3/2$ has a different equality structure. In the scalar finite-rank problem, Frank, Gontier and Lewin [3, Theorem 5] identify the optimizers as KdV multisolitons. In particular, several bound states can occur in a single scalar channel at that endpoint. The conclusion of Theorem 1.1, one bound state per nonzero channel, uses the strict inequality $r > 1$ and is asserted only for the open interval.

1.2 Quantitative defects and fixed channels

The proof begins with the finite-interval matrix action method of [11]. For orthonormal real trial functions approximating finitely many negative eigenfunctions, place the functions in the rows of $U(x)$ and choose a lower triangular matrix C . The rows of $A = CU$ remain in the successive trial spans. If K is the diagonal matrix of positive square roots of the trial binding energies, the action involves A' , $K^2 AA^T$, and a power of the density AA^T . A symmetric matrix path $B(x)$ connects K to $-K$ and gives a lower bound for that action.

For equality, the lower bound must retain more information. Besides the square $\|A' - BA\|_{\text{HS}}^2$, we retain a term controlling

$$[B, K^2] \quad \text{and} \quad K^2 - B^2 - (AA^T)^r.$$

Its coefficient depends on the largest binding energy and the exponent, but not on the number of trial functions. This uniformity is the first essential point: an extremal potential is not known in advance to have finitely many negative eigenvalues. The integrated trace identity of the matrix companion supplies the two nonnegative squares from which this estimate follows.

The second point is compactness with a fixed number of columns. As the trial list grows, A has more rows but still only m columns. Its density therefore has rank at most m . The retained defect bounds the tail rows by the small binding energies in that tail. This yields strong pointwise convergence even for an infinite eigenvalue list and gives exact algebraic relations in the limit. Those relations separate the rows with different energies. A differential inequality then makes their row spaces independent of position; their mutual orthogonality leaves at most m nonzero directions. On each resulting finite block, a matrix Riccati equation has a fixed diagonalizing basis and determines the profiles in (1.4).

Section 2 gives the spectral and convexity preliminaries. Section 3 proves the quantitative action estimate, and Section 4 recovers the sharp spectral bound. Section 5 passes to exact relations for an equality case. Section 6 proves that the channels are fixed, solves the Riccati equation, and treats complex potentials and the converse. Appendices A and B prove the trace identity, regularized field construction, and connecting-path theorem used by the action estimate.

1.3 Notation

Throughout the proof,

$$\sigma = \gamma - \frac{1}{2} \in (0, 1), \quad p = 1 + \sigma, \quad r = \sigma^{-1}, \quad q = r + 1, \quad (1.5)$$

and

$$D_\sigma = \frac{\sigma^\sigma}{(1 + \sigma)^{1+\sigma}}, \quad b_\sigma = \int_{-1}^1 (1 - s^2)^\sigma ds, \quad C_\gamma = \frac{D_\sigma}{b_\sigma}. \quad (1.6)$$

The last identity follows from the beta integral. We use the operator norm $\|\cdot\|_{\text{op}}$ and the Hilbert–Schmidt norm $\|A\|_{\text{HS}}^2 = \text{tr}(A^*A)$. The symbol T denotes transpose of a real matrix, and Sym_n is the space of real symmetric $n \times n$ matrices with inner product $\text{tr}(BH)$. The commutator is $[B, H] = BH - HB$. The identity matrix, or identity operator, has the size determined by its context.

2 Spectral and convexity preliminaries

Two elementary facts let us work with measurable potentials and identify the equality condition in the potential estimate. The first is the form argument used in the matrix inequality [11, Lemma 7.1]; we give its proof to fix the precise regularity available below.

Lemma 2.1. *Let $p > 1$ and let $W : \mathbb{R} \rightarrow \mathbb{C}^{m \times m}$ be measurable, Hermitian and positive semidefinite, with $\int_{\mathbb{R}} \text{tr}(W^p) < \infty$. The form h_W on $H^1(\mathbb{R}; \mathbb{C}^m)$ is closed and bounded below. Multiplication by $W^{1/2}$ is compact from H^1 to L^2 . The negative spectrum of H_W consists of isolated eigenvalues of finite multiplicity, with possible accumulation only at zero. Its eigenfunctions belong to H^1 and satisfy $-\psi'' - W\psi = \lambda\psi$ distributionally.*

Proof. Set $w = \|W\|_{\text{op}}$. Then $w^p \leq \text{tr}(W^p)$, so $w \in L^p$. The one-dimensional Sobolev estimate, applied to the norm of a vector function, gives $\|\psi\|_\infty^2 \leq 2\|\psi\|_2\|\psi'\|_2$. Hölder's inequality, interpolation and Young's inequality imply

$$\int_{\mathbb{R}} \langle \psi, W\psi \rangle \leq 2^{1/p} \|w\|_p \|\psi'\|_2^{1/p} \|\psi\|_2^{2-1/p} \leq \varepsilon \|\psi'\|_2^2 + C_\varepsilon \|\psi\|_2^2. \quad (2.1)$$

A sufficiently shifted form norm is therefore equivalent to the H^1 norm. This proves closedness and semiboundedness, and polarization gives continuity of the potential form on H^1 .

Write $T\psi = W^{1/2}\psi$. The truncated multiplier $T_{R,L} = \mathbf{1}_{\{|x| \leq R, w \leq L\}} T$ is compact by the compact embedding of $H^1([-R, R])$ into $L^2([-R, R])$. Moreover,

$$\|(T - T_{R,L})\psi\|_2^2 \leq C \|w \mathbf{1}_{\{|x| > R\} \cup \{w > L\}}\|_p \|\psi\|_{H^1}^2.$$

The coefficient tends to zero, so T is compact. For each $a > 0$, an infinite-dimensional spectral subspace in $(-\infty, -a]$ would contain an L^2 -orthonormal sequence bounded in H^1 by (2.1). A subsequence converges weakly to zero in H^1 , hence its images under T converge strongly to zero. This contradicts $-a \geq h_W[\psi] \geq -\|T\psi\|_2^2$ for unit vectors in that subspace. Thus every such subspace is finite-dimensional. Finally, the weak form equation tested against smooth compactly supported functions is the stated distributional equation; its potential term is locally integrable because $W \in L_{\text{loc}}^p$ and ψ is locally bounded. $\square$

Lemma 2.2 (Trace Young inequality and its equality condition). *For positive semidefinite matrices W, Z of the same finite size, set*

$$Y(W, Z) = D_\sigma \operatorname{tr}(W^p) - \operatorname{tr}(WZ) + \operatorname{tr}(Z^q). \quad (2.2)$$

Then $Y(W, Z) \geq 0$, with equality if and only if $W = qZ^r$. Also,

$$Y(W, Z) \geq \frac{1}{2} \operatorname{tr}(Z^q) - (2^{p-1} - 1) D_\sigma \operatorname{tr}(W^p). \quad (2.3)$$

Proof. Scalar maximization over $z \geq 0$ gives $yz - z^q \leq D_\sigma y^p$ for $y \geq 0$, with equality precisely when $y = qz^r$. In an orthonormal eigenbasis (v_j) of Z , let z_j be its eigenvalues and put $y_j = \langle v_j, Wv_j \rangle$. Strict convexity of t^p on $[0, \infty)$ gives

$$\operatorname{tr}(WZ) - \operatorname{tr}(Z^q) \leq D_\sigma \sum_j y_j^p \leq D_\sigma \sum_j \langle v_j, W^p v_j \rangle.$$

Equality in the second inequality holds exactly when each v_j belongs to an eigenspace of W . Equality in the first then requires $Wv_j = qz_j^r v_j$ for every j . This proves both directions of the equality assertion, including zero eigenvalues. Applying the same inequality to $2W, Z$ gives $\operatorname{tr}(WZ) \leq \frac{1}{2} \operatorname{tr}(Z^q) + 2^{p-1} D_\sigma \operatorname{tr}(W^p)$, which is (2.3). $\square$

3 An action estimate that retains the equality defects

The finite-interval argument for the sharp inequality can also measure failure of its equality relations. We retain the first-order square and a quantitative part of the matrix trace deficit. The coefficient of this second defect will be independent of the number of rows. This uniformity allows us to exhaust an eigenvalue list before knowing that it is finite.

The construction adapts the field, connecting paths, and action identity of [11, Sections 3–6]. Their analytic foundations are proved in Appendices A and B; the additional estimate below extracts operator-norm rigidity from the trace deficit.

Fix positive integers n, d , a bounded interval $\Omega = (x_-, x_+)$, and $K = \operatorname{diag}(\ell_1, \dots, \ell_n)$ with $0 < \ell_i \leq \kappa$. For $A \in H_0^1(\Omega; \mathbb{R}^{n \times d})$, set $\rho = AA^\top$ and define

$$\mathcal{E}_K(A) = \int_\Omega (\|A'\|_{\text{HS}}^2 + \operatorname{tr}(K^2 \rho) - \operatorname{tr}(\rho^q)) \, dx. \quad (3.1)$$

For pointwise matrix values $A \in \mathbb{R}^{n \times d}$ and $B \in \operatorname{Sym}_n$, put

$$\Phi_K(A, B) = \min \left\{ 1, \|[B, K^2]\|_{\text{op}}^2 + \|K^2 - B^2 - (AA^\top)^r\|_{\text{op}}^2 \right\}. \quad (3.2)$$

Thus $\Phi_K = 0$ expresses two algebraic relations, while $A' - BA = 0$ is a first-order differential relation.

Theorem 3.1 (Quantitative finite-interval action estimate). *Let $0 < \sigma < 1$ and $\kappa > 0$. There is a constant $c_* = c_*(\sigma, \kappa) > 0$ with the following property. For n, d, Ω, K as above, suppose $U \in H_0^1(\Omega; \mathbb{R}^{n \times d})$ satisfies*

$$\int_\Omega UU^\top \, dx = I_n.$$

For every $\eta > 0$ there exist a real lower triangular matrix $C \in \mathbb{R}^{n \times n}$ and a path $B \in C^1(\overline{\Omega}; \operatorname{Sym}_n)$ such that

$$B(x_-) = K, \quad B(x_+) = -K, \quad \sup_{x \in \overline{\Omega}} \|B(x)\|_{\text{op}} \leq \kappa,$$

and, for $A = CU$,

$$\mathcal{E}_K(A) \geq b_\sigma \sum_{i=1}^n \ell_i^{2\gamma} + \int_{\Omega} \left(\|A' - BA\|_{\text{HS}}^2 + c_* \Phi_K(A, B) \right) dx - \eta. \quad (3.3)$$

In particular, c_* is independent of n, d, Ω, U , and the smallest diagonal entry of K .

3.1 Coercivity at the matrix-field constraint

The algebraic relations $\Phi_K(A, B) = 0$ require $[B, K^2] = 0$ and $\rho = (K^2 - B^2)^\sigma$. For $0 < \delta \leq 1$, we use a locally Lipschitz field $M_\delta : \text{Sym}_n \rightarrow \text{Sym}_n$ extending the commuting rule

$$M_\delta(B) = (K^2 - B^2 + \delta I)^\sigma - \delta^\sigma I \quad \text{if } [B, K^2] = 0 \text{ and } K^2 - B^2 \geq 0.$$

Thus the constraint $M_\delta(B) = \rho$ is a regularized form of the algebraic equality relation in this case. The extension to arbitrary symmetric B has a C^1 primitive $J_\delta : \text{Sym}_n \rightarrow \mathbb{R}$ satisfying

$$dJ_\delta(B)[H] = -\text{tr}(M_\delta(B)H). \quad (3.4)$$

Proposition B.1 constructs these maps and proves their required properties. Along a path B , the derivative identity converts $-\text{tr}(M_\delta(B)B')$ into the total derivative of $J_\delta(B)$, which will supply the boundary term in the action identity. Its endpoint difference is explicit:

$$\begin{aligned} J_\delta(-K) - J_\delta(K) &= \sum_{i=1}^n \int_{-\ell_i}^{\ell_i} ((\ell_i^2 - s^2 + \delta)^\sigma - \delta^\sigma) ds \\ &\longrightarrow b_\sigma \sum_{i=1}^n \ell_i^{2\gamma} \quad (\delta \downarrow 0). \end{aligned} \quad (3.5)$$

The limit follows by dominated convergence and $2\sigma + 1 = 2\gamma$.

Define the scalar function

$$F(A, B) = \text{tr}(\rho^q) - \text{tr}((K^2 - B^2)\rho), \quad \rho = AA^\top. \quad (3.6)$$

The following lemma bounds this nonlinear term while keeping the algebraic defect. The constraint in its hypothesis will later be enforced by a penalty.

Lemma 3.2. *The constant $c_*(\sigma, \kappa) > 0$ can be chosen so that, whenever $0 < \delta \leq 1$, $\|B\|_{\text{op}} \leq \kappa$, and $M_\delta(B) = AA^\top$, one has*

$$F(A, B) + c_* \Phi_K(A, B) \leq E_{n, \delta}, \quad E_{n, \delta} = n(\delta \kappa^{2\sigma} + \delta^\sigma(\kappa^2 + \delta)) + \delta^{2\sigma}. \quad (3.7)$$

Proof. Write $\rho = M_\delta(B) \geq 0$. Proposition B.2 gives

$$Q_s := s^2 I - 2sB + K^2 \geq 0 \quad (s \in \mathbb{R}), \quad 0 \leq \rho \leq \kappa^{2\sigma} I. \quad (3.8)$$

Put $V = K^2 - B^2 + \delta I$ and $\beta = \sigma - \frac{1}{2}$. With the positive normalization constant c_σ specified in (A.1), define

$$\begin{aligned} R(t) &= \frac{1}{\pi} \int_{\mathbb{R}} (Q_s + (\delta + t)I)^{-1} ds, \quad t > 0, \\ N &= c_\sigma \int_0^\infty t^\beta (t^{-1/2} I - R(t)) dt = \rho + \delta^\sigma I. \end{aligned} \quad (3.9)$$

The last identity is also part of Proposition B.2. Since $(sI - B)^2 + V = Q_s + \delta I \geq \delta I$ and $N \geq 0$, Theorem A.1 applies. It supplies $L(t) = R(t)^{-1} > 0$ and Hermitian matrices $Z(t)$ satisfying

$$V + tI = L^2 + Z^2 + i[B, Z], \quad LZ + ZL + i[B, L] = 0, \quad (3.10)$$

and the exact identity

$$\begin{aligned} \Delta &:= \operatorname{tr}(NV) - \operatorname{tr}(N^q) \\ &= c_\sigma \int_0^\infty t^\beta \left(\operatorname{tr}(R(t)Z(t)^2) + q \|R(t)^{-1/2} - R(t)^{1/2}(tI + N^r)^{1/2}\|_{\text{HS}}^2 \right) dt \geq 0. \end{aligned} \quad (3.11)$$

The matrices in (3.10) may be complex Hermitian even though A and B are real; i denotes the imaginary unit.

We extract a dimension-independent bound from the part of the integral with $1 \leq t \leq 2$. On this interval there are constants $a_\kappa, A_\kappa > 0$ such that

$$a_\kappa I \leq R(t) \leq A_\kappa I. \quad (3.12)$$

For the upper bound, use $Q_s \geq 0$ on a bounded interval in s , and $Q_s \geq \frac{1}{2}s^2 I$ for $|s| \geq \max\{1, 4\kappa\}$. For the lower bound, restrict the integral to $|s| \leq 1$, where $Q_s + (\delta + t)I \leq (4 + 2\kappa + \kappa^2)I$. These estimates depend on κ alone. Moreover $\|N\|_{\text{op}} \leq \kappa^{2\sigma} + 1$, so $H_t = (tI + N^r)^{1/2}$ and $L(t)$ are bounded in operator norm by constants depending only on σ, κ .

The two integrands in (3.11) control $\|Z(t)\|_{\text{HS}}^2$ and $\|L(t) - H_t\|_{\text{HS}}^2$: indeed $\operatorname{tr}(RZ^2) = \operatorname{tr}(ZRZ)$ and

$$R^{-1/2} - R^{1/2}H_t = R^{1/2}(L - H_t).$$

It follows by averaging over $[1, 2]$ that at some t in this interval,

$$\|Z(t)\|_{\text{op}} + \|L(t) - H_t\|_{\text{op}} \leq C_1 \sqrt{\Delta}, \quad (3.13)$$

where C_1 depends only on σ, κ . When $\Delta = 0$, the same conclusion follows by taking a point where both nonnegative integrands vanish.

Suppose first that $\Delta \leq 1$, and use this value of t . The first identity in (3.10) gives

$$V - N^r = (L^2 - H_t^2) + Z^2 + i[B, Z].$$

The boundedness of L, H_t, B , together with (3.13), bounds the norm of this expression by $C_2 \sqrt{\Delta}$. To control the commutator, the second identity in (3.10) first yields $\|[B, L]\|_{\text{op}} \leq 2\|L\|_{\text{op}}\|Z\|_{\text{op}}$. Then

$$[B, V] = [B, L^2] + [B, Z^2] + i[B, [B, Z]]$$

has the same bound. Thus

$$\|V - N^r\|_{\text{op}} + \|[B, V]\|_{\text{op}} \leq C_3 \sqrt{\Delta}. \quad (3.14)$$

Here and below the constants in this proof depend only on σ, κ .

The matrices $N = \rho + \delta^\sigma I$ and ρ commute. Because $r > 1$, the scalar function $y \mapsto y^r$ is Lipschitz on $[0, \kappa^{2\sigma} + 1]$. Consequently

$$\|N^r - \rho^r\|_{\text{op}} \leq C_4 \delta^\sigma.$$

Since $[B, V] = [B, K^2]$ and $\delta \leq \delta^\sigma$, (3.14) implies

$$\|[B, K^2]\|_{\text{op}}^2 + \|K^2 - B^2 - \rho^r\|_{\text{op}}^2 \leq C_5(\Delta + \delta^{2\sigma}).$$

Choose $c_* > 0$ sufficiently small, also with $c_* \leq 1$. For $\Delta \leq 1$ the preceding estimate gives

$$c_* \Phi_K(A, B) \leq \Delta + \delta^{2\sigma}. \quad (3.15)$$

For $\Delta > 1$ this follows directly from $\Phi_K \leq 1$. This proves the required uniform control of the algebraic defect.

It remains to relate Δ to F . Expanding N and V gives

$$F(A, B) = -\Delta + \text{tr}(\rho^q) - \text{tr}(N^q) + \delta \text{tr} \rho + \delta^\sigma \text{tr} V.$$

The first two powers are ordered because $N = \rho + \delta^\sigma I$. Also $\text{tr} \rho \leq n\kappa^{2\sigma}$ and $V \leq (\kappa^2 + \delta)I$. Adding (3.15) now proves (3.7). $\square$

3.2 The action identity and the penalty limit

Proof of Theorem 3.1. Fix $0 < \delta \leq 1$. Proposition B.3 supplies, for every $0 < \varepsilon \leq 1$, a lower triangular matrix C_ε and a path B_ε with the prescribed endpoints and operator-norm bound such that, for $A_\varepsilon = C_\varepsilon U$,

$$\varepsilon B'_\varepsilon = M_\delta(B_\varepsilon) - A_\varepsilon A_\varepsilon^\top. \quad (3.16)$$

For fixed δ, K, Ω, U , the matrices C_ε are bounded uniformly in $0 < \varepsilon \leq 1$.

Temporarily suppress the subscript ε and write

$$u_\delta(A, B) = \|M_\delta(B) - AA^\top\|_{\text{HS}}^2.$$

Since $A \in H_0^1$ and $B \in C^1$, the product and chain rules for absolutely continuous functions, (3.4), and (3.16) yield almost everywhere

$$\begin{aligned} (J_\delta(B) + \text{tr}(A^\top BA))' &= 2 \text{tr}((A')^\top BA) - \text{tr}((M_\delta(B) - AA^\top)B') \\ &= 2 \text{tr}((A')^\top BA) - \varepsilon^{-1} u_\delta(A, B). \end{aligned}$$

Complete the square in the action and integrate this identity. Because A vanishes at both endpoints, the result is the exact formula

$$\begin{aligned} \mathcal{E}_K(A) &= J_\delta(-K) - J_\delta(K) \\ &\quad + \int_\Omega \left(\|A' - BA\|_{\text{HS}}^2 - F(A, B) + \varepsilon^{-1} u_\delta(A, B) \right) dx. \end{aligned} \quad (3.17)$$

We now use the penalty to pass from the constraint estimate in Lemma 3.2 to all matrix values on these paths. The continuous representative of U is bounded on $\bar{\Omega}$. The uniform bound on C_ε and the bound $\|B_\varepsilon\|_{\text{op}} \leq \kappa$ therefore place all pairs $(A_\varepsilon(x), B_\varepsilon(x))$ in one compact set $\mathcal{K}_\delta \subset \mathbb{R}^{n \times d} \times \text{Sym}_n$ on which $\|B\|_{\text{op}} \leq \kappa$. On this set F , Φ_K , and u_δ are continuous. Lemma 3.2 implies

$$\limsup_{\varepsilon \downarrow 0} \sup_{(A, B) \in \mathcal{K}_\delta} (F(A, B) + c_* \Phi_K(A, B) - \varepsilon^{-1} u_\delta(A, B)) \leq E_{n, \delta}. \quad (3.18)$$

Indeed, if the left side exceeded $E_{n, \delta}$, some sequence $\varepsilon_j \downarrow 0$ and corresponding pairs in the compact set would make the expression at least $E_{n, \delta} + \tau$ for some $\tau > 0$. Boundedness of $F + c_* \Phi_K$ would

force $u_\delta = O(\varepsilon_j)$ along that sequence. A convergent subsequence would then give a limiting pair with $u_\delta = 0$ and $F + c_*\Phi_K \geq E_{n,\delta} + \tau$, contradicting Lemma 3.2.

For any $\tau > 0$, choose ε small enough that the supremum in (3.18) is at most $E_{n,\delta} + \tau$. Substitution into (3.17) gives

$$\begin{aligned} \mathcal{E}_K(A_\varepsilon) &\geq J_\delta(-K) - J_\delta(K) - |\Omega|(E_{n,\delta} + \tau) \\ &\quad + \int_\Omega \left(\|A'_\varepsilon - B_\varepsilon A_\varepsilon\|_{\text{HS}}^2 + c_*\Phi_K(A_\varepsilon, B_\varepsilon) \right) dx. \end{aligned}$$

Finally, first choose $\delta > 0$ so small that the endpoint error in (3.5) and $|\Omega|E_{n,\delta}$ together are less than $\eta/2$. Next choose $\tau > 0$ with $|\Omega|\tau < \eta/2$, and then choose the required ε . These choices prove (3.3). All compactness arguments were made after fixing δ ; no bound uniform in δ is needed. $\square$

4 Recovering the sharp spectral bound

The action estimate implies the sharp inequality by the spectral passage of [11, Section 7]. We record it before studying equality, so that the finite spectral moment used below has already been established.

Proposition 4.1. *Every potential in Theorem 1.1 satisfies*

$$\text{Tr}(H_W)_-^\gamma \leq \frac{D_\sigma}{b_\sigma} \int_{\mathbb{R}} \text{tr}(W^p).$$

Proof. First suppose that W is real symmetric. Choose any finite initial list of negative eigenvalues $-k_1^2, \dots, -k_n^2$, with $k_1 \geq \dots \geq k_n > 0$, and real orthonormal eigenfunctions φ_i . For $0 < \xi < k_n^2$ put $\ell_i = (k_i^2 - \xi)^{1/2}$ and $K = \text{diag}(\ell_1, \dots, \ell_n)$. There are real smooth compactly supported orthonormal functions u_i with form matrix

$$(h_W[u_i, u_j])_{i,j=1}^n \leq -K^2. \quad (4.1)$$

Indeed, approximate the φ_i in H^1 , then multiply the family by the inverse square root of its Gram matrix. The Gram matrix tends to I_n , and the form matrix tends to $-\text{diag}(k_i^2)$ by Lemma 2.1; an operator norm error below ξ gives (4.1).

Let U have rows u_i , and choose a bounded interval containing their supports. For every lower triangular C , the i th row a_i of $A = CU$ satisfies

$$h_W[a_i] + \ell_i^2 \|a_i\|_2^2 \leq \sum_{l \leq i} C_{il}^2 (\ell_i^2 - \ell_l^2) \leq 0.$$

Set $Z = A^T A$. The matrices AA^T and Z have the same nonzero eigenvalues, with multiplicity. Summing the preceding bound and applying Lemma 2.2 yields

$$\mathcal{E}_K(A) \leq \int_{\mathbb{R}} \{\text{tr}(WZ) - \text{tr}(Z^q)\} \leq D_\sigma \int_{\mathbb{R}} \text{tr}(W^p).$$

Theorem 3.1, with its nonnegative defects discarded, gives $b_\sigma \sum_i \ell_i^{2\gamma} \leq D_\sigma \int \text{tr}(W^p) + \eta$ for every $\eta > 0$. Let $\eta \downarrow 0$, then $\xi \downarrow 0$, and finally exhaust the negative eigenvalues. The case of an empty negative spectrum is immediate.

For complex Hermitian $W = E + iF$, with E, F real, its realification is

$$\widetilde{W} = \begin{pmatrix} E & -F \\ F & E \end{pmatrix}. \quad (4.2)$$

The constant unitary matrix $2^{-1/2} \begin{pmatrix} I & iI \\ I & -iI \end{pmatrix}$ conjugates $\widetilde{W}$ to $W \oplus \overline{W}$. Hence $\widetilde{W}$ is real symmetric and positive semidefinite, $\text{tr}(\widetilde{W}^p) = 2 \text{tr}(W^p)$, and $\text{Tr}(H_{\widetilde{W}}^\gamma) = 2 \text{Tr}(H_W^\gamma)$. Here conjugation identifies the spectra of H_W and $H_{\overline{W}}$. The real inequality, divided by two, proves the result. $\square$

5 From equality to pointwise matrix relations

We now apply Theorem 3.1 to finite collections of negative eigenfunctions. Equality forces each of its nonnegative errors to vanish. The main issue is that the number of eigenfunctions may grow without bound, whereas the potential has only m channels. We will use this fixed number of columns to control the tails of the resulting matrices.

Throughout this section, W is real symmetric, $W \geq 0$ almost everywhere, and $0 < \int_{\mathbb{R}} \text{tr}(W^p) < \infty$. Suppose that equality holds, in the form

$$b_\sigma \sum_{i \in \mathcal{I}} k_i^{2\gamma} = D_\sigma \int_{\mathbb{R}} \text{tr}(W^p) dx, \quad (5.1)$$

where $-k_i^2$ are the negative eigenvalues of H_W , repeated according to multiplicity, and

$$k_1 \geq k_2 \geq \dots > 0.$$

By Lemma 2.1, the index set $\mathcal{I}$ is a nonempty finite initial segment of $\mathbb{N}$ or all of $\mathbb{N}$, each eigenvalue has finite multiplicity, and $k_i \rightarrow 0$ in the infinite case. Fix real orthonormal eigenfunctions $\varphi_i \in H^1(\mathbb{R}; \mathbb{R}^m)$ and regard them as row vectors. Write

$$\mathcal{H} = \ell^2(\mathcal{I}; \mathbb{R}), \quad \Lambda = \text{diag}(k_i : i \in \mathcal{I}), \quad \kappa = k_1.$$

For operators between real Hilbert spaces, the superscript T will also denote the adjoint.

Proposition 5.1 (Pointwise relations at equality). *Under these assumptions, there are rows $a_i \in H^1(\mathbb{R}; \mathbb{R}^m)$, $i \in \mathcal{I}$, each a linear combination of eigenfunctions with eigenvalue $-k_i^2$, such that*

$$-a_i'' - a_i W = -k_i^2 a_i \quad \text{in distributions.} \quad (5.2)$$

The rows a_i are allowed to vanish. On a common set of full measure, they define a Hilbert–Schmidt operator $A(x) : \mathbb{R}^m \rightarrow \mathcal{H}$, and there exists a bounded self-adjoint operator B on $\mathcal{H}$ with $\|B\|_{\text{op}} \leq \kappa$ such that

$$[B, \Lambda^2] = 0, \quad \Lambda^2 - B^2 = (A A^T)^r, \quad a_i'(x) = (B A)_{i,\cdot} \quad (i \in \mathcal{I}). \quad (5.3)$$

Moreover,

$$W(x) = q(A(x)^T A(x))^r \quad \text{almost everywhere.} \quad (5.4)$$

The operator B is asserted to exist separately at each such point x ; no measurable choice is required.

We divide the proof into the construction of a sequence whose errors vanish, convergence of every fixed row, and control of the remaining rows.

5.1 Finite spectral approximations and vanishing errors

Let $n_j = j$ when $\mathcal{I} = \mathbb{N}$, and let $n_j = |\mathcal{I}|$ when $\mathcal{I}$ is finite. Choose $0 < \xi_j < 1/j$ so small that

$$\ell_{j,i} := (k_i^2 - \xi_j)^{1/2} > 0 \quad (1 \leq i \leq n_j), \quad \sum_{i \leq n_j} (k_i^{2\gamma} - \ell_{j,i}^{2\gamma}) < 1/j. \quad (5.5)$$

Set $K_j = \text{diag}(\ell_{j,1}, \dots, \ell_{j,n_j})$. The same shift ξ_j is used for every eigenvalue: this leaves the differences of their squares unchanged.

The approximation leading to (4.1) can be made arbitrarily accurate in H^1 . Apply it to the first n_j eigenfunctions, choosing the H^1 errors below $1/j$ and the form-matrix error below ξ_j . This gives real, smooth, compactly supported functions $u_{j,1}, \dots, u_{j,n_j}$, orthonormal in $L^2(\mathbb{R}; \mathbb{R}^m)$, such that

$$\|u_{j,i} - \varphi_i\|_{H^1} < 1/j, \quad (h_W[u_{j,i}, u_{j,l}])_{i,l=1}^{n_j} \leq -K_j^2. \quad (5.6)$$

Here $h_W[\cdot, \cdot]$ is the polarized quadratic form, and the second inequality is the ordering of real symmetric matrices.

Let U_j be the matrix with rows $u_{j,i}$, and choose a bounded interval Ω_j containing their supports in its interior. Apply Theorem 3.1 with $K = K_j$, $d = m$, $\kappa = k_1$, and $\eta = 1/j$. It gives a lower triangular matrix C_j , a symmetric path B_j with $\|B_j\|_{\text{op}} \leq \kappa$, and $A_j = C_j U_j$. Extend A_j by zero outside Ω_j and extend B_j by K_j to the left and by $-K_j$ to the right. All the errors in that theorem vanish outside Ω_j , so their integrals may be taken over $\mathbb{R}$.

The lower triangular form of C_j gives a useful additional error. Put $Z_j = A_j^T A_j$, an $m \times m$ positive semidefinite matrix. By (5.6),

$$\mathcal{E}_{K_j}(A_j) \leq \int_{\mathbb{R}} (\text{tr}(W Z_j) - \text{tr}(Z_j^q)) dx - T_j, \quad T_j = \sum_{l \leq i \leq n_j} (C_j)_{il}^2 (k_l^2 - k_i^2) \geq 0. \quad (5.7)$$

Indeed, the i th row of A_j is $\sum_{l \leq i} (C_j)_{il} u_{j,l}$. Its quadratic form, plus $\ell_{j,i}^2$ times its squared L^2 norm, is at most

$$\sum_{l \leq i} (C_j)_{il}^2 (\ell_{j,i}^2 - \ell_{j,l}^2) = - \sum_{l \leq i} (C_j)_{il}^2 (k_l^2 - k_i^2).$$

Summing over i proves (5.7), since $A_j A_j^T$ and Z_j have the same nonzero eigenvalues, with multiplicity. The ordering of the k_i makes every summand of T_j nonnegative.

Recall the nonnegative Young deficit from Lemma 2.2:

$$Y(W, Z) = D_\sigma \text{tr}(W^p) - \text{tr}(WZ) + \text{tr}(Z^q) \geq 0, \quad Y(W, Z) = 0 \iff W = qZ^r. \quad (5.8)$$

Combining Theorem 3.1, (5.1), and (5.7) yields

$$0 \leq T_j + \int_{\mathbb{R}} \left(Y(W, Z_j) + \|A'_j - B_j A_j\|_{\text{HS}}^2 + c_* \Phi_{K_j}(A_j, B_j) \right) dx \leq d_j, \quad (5.9)$$

$$d_j := b_\sigma \left(\sum_{i \in \mathcal{I}} k_i^{2\gamma} - \sum_{i \leq n_j} \ell_{j,i}^{2\gamma} \right) + 1/j \longrightarrow 0.$$

Here $c_* > 0$ is fixed: its independence of n_j is essential. Thus all three integral errors tend to zero, as does T_j .

We will also need a uniform bound on A_j . The coercive estimate (2.3) gives

$$Y(W, Z_j) \geq \frac{1}{2} \text{tr}(Z_j^q) - (2^{p-1} - 1) D_\sigma \text{tr}(W^p). \quad (5.10)$$

Since Z_j has size m , scalar Hölder's inequality gives

$$\|A_j\|_{\text{HS}}^{2q} = (\text{tr } Z_j)^q \leq m^{q-1} \text{tr}(Z_j^q).$$

Equations (5.9) and (5.10) therefore bound A_j uniformly in $L^{2q}(\mathbb{R})$ with the Hilbert–Schmidt norm, independently of its number of rows.

5.2 Convergence of each fixed row

For fixed $i \in \mathcal{I}$, the coefficients $(C_j)_{il}$, $l \leq i$, are bounded as $j \rightarrow \infty$. To see this, choose a bounded interval J_i so large that the restricted Gram matrix

$$\left(\int_{J_i} \varphi_l(x) \varphi_s(x)^T dx \right)_{l,s=1}^i \geq \frac{3}{4} I_i,$$

where I_i denotes the $i \times i$ identity matrix. Such an interval exists because the full Gram matrix is the identity. By (5.6), the corresponding restricted Gram matrix of the $u_{j,l}$ is at least one half of the identity for all sufficiently large j . Hence

$$\frac{1}{2} \sum_{l \leq i} (C_j)_{il}^2 \leq \int_{J_i} |(A_j)_{i,\cdot}|^2 dx \leq |J_i|^{1-1/q} \left(\int_{J_i} \|A_j\|_{\text{HS}}^{2q} dx \right)^{1/q}.$$

The last expression is uniformly bounded. Lower triangularity is used here to keep the number of coefficients in a fixed row finite.

A diagonal subsequence now makes every coefficient $(C_j)_{il}$ converge to a limit c_{il} , for $l \leq i$. From $T_j \rightarrow 0$ and (5.7),

$$c_{il} = 0 \quad \text{whenever } k_l > k_i. \quad (5.11)$$

Define

$$a_i = \sum_{l \leq i} c_{il} \varphi_l.$$

The sum is finite, and (5.6) implies

$$(A_j)_{i,\cdot} \longrightarrow a_i \quad \text{in } H^1(\mathbb{R}; \mathbb{R}^m) \quad \text{for every fixed } i. \quad (5.12)$$

The ordering of the eigenvalues and (5.11) show that only eigenfunctions with eigenvalue $-k_i^2$ occur in a_i . This proves (5.2), including the possibility $a_i = 0$. The eigenfunction equations hold distributionally by testing the form equations against smooth compactly supported functions; the potential is locally integrable and the rows are locally bounded.

Choose the continuous, locally absolutely continuous representatives of these rows. The Sobolev embedding $H^1(\mathbb{R}) \subset C_b(\mathbb{R})$ shows that (5.12) also gives pointwise convergence at every x . Passing to a further subsequence, we may arrange that the integrals of the nonnegative integrand in (5.9) are summable, and that, for every fixed i , the squared L^2 errors of the derivatives in (5.12) are summable. A diagonal choice achieves both conditions, by making the first j available derivative errors and the integral error less than 2^{-j} along the chosen subsequence. Consequently, on a common set of full measure,

$$\begin{aligned} Y(W, Z_j) &\rightarrow 0, & \|A'_j - B_j A_j\|_{\text{HS}} &\rightarrow 0, & \Phi_{K_j}(A_j, B_j) &\rightarrow 0, \\ (A'_j)_{i,\cdot}(x) &\rightarrow a'_i(x) & \text{for every } i \in \mathcal{I}. \end{aligned} \quad (5.13)$$

We also exclude the null set where W is not a finite positive semidefinite matrix. It remains to show that convergence of the individual rows is strong enough to pass to the products in these relations.

5.3 Tail control in the row-index space

Pad A_j , B_j , and K_j with zeros in the unused coordinates of $\mathcal{H}$. Thus $A_j : \mathbb{R}^m \rightarrow \mathcal{H}$, and B_j, K_j are self-adjoint finite-rank operators on $\mathcal{H}$. Since the omitted k_i tend to zero,

$$\|K_j^2 - \Lambda^2\|_{\text{op}} \leq \max\{\xi_j, \sup_{i>n_j} k_i^2\} \longrightarrow 0, \quad (5.14)$$

where a supremum over no indices is zero.

Fix a point x in the full-measure set from (5.13) and suppress x for the moment. Write $\rho_j = A_j A_j^T$. The definition of Φ and its convergence to zero give

$$\|[B_j, K_j^2]\|_{\text{op}} \rightarrow 0, \quad e_j := \|K_j^2 - B_j^2 - \rho_j^r\|_{\text{op}} \rightarrow 0. \quad (5.15)$$

For $h \geq 1$, let P_h be the orthogonal projection onto the first h coordinates of $\mathcal{H}$, with all coordinates included if h exceeds $|\mathcal{I}|$. Set $Q_h = I - P_h$ and $s_h = \sup_{i>h} k_i^2$, again with the empty supremum equal to zero. Then $s_h \rightarrow 0$. For every unit vector v in the range of Q_h ,

$$\|B_j v\|^2 + \langle v, \rho_j^r v \rangle \leq s_h + e_j. \quad (5.16)$$

Both terms on the left are nonnegative. Jensen's inequality for the spectral measure of ρ_j , using $r > 1$, implies

$$\langle v, \rho_j^r v \rangle \leq \langle v, \rho_j^r v \rangle.$$

It follows that $\|Q_h \rho_j Q_h\|_{\text{op}} \leq (s_h + e_j)^{1/r}$. Although $\mathcal{H}$ may be infinite dimensional, $Q_h \rho_j Q_h$ has rank at most m , because A_j has exactly m columns. Its trace is therefore at most m times its operator norm. We obtain the two tail bounds

$$\boxed{\|Q_h A_j\|_{\text{HS}}^2 \leq m(s_h + e_j)^{1/r}, \quad \|B_j Q_h\|_{\text{op}} \leq (s_h + e_j)^{1/2}.} \quad (5.17)$$

The first estimate converts decay of the eigenvalue parameters into decay of the total mass of all omitted rows. Its factor is the fixed physical dimension m , rather than the number of retained eigenfunctions; Figure 1 depicts this distinction.

For each fixed h , the first h rows of A_j converge by (5.12). The first bound in (5.17) makes the remaining rows uniformly small by choosing h large and then j sufficiently large. Thus A_j is Cauchy in Hilbert–Schmidt norm and converges to an operator $A : \mathbb{R}^m \rightarrow \mathcal{H}$ whose rows are precisely $a_i(x)$:

$$A_j \longrightarrow A \quad \text{in Hilbert–Schmidt norm.} \quad (5.18)$$

This argument applies to the whole subsequence already selected and does not require a further choice depending on x .

For B_j we need only pointwise compactness. Its finite compressions $P_h B_j P_h$ are bounded in finite-dimensional spaces, so a diagonal subsequence makes them converge for every h . Self-adjointness and the second estimate in (5.17) give

$$\|B_j - P_h B_j P_h\|_{\text{op}} \leq 2\|B_j Q_h\|_{\text{op}} \leq 2(s_h + e_j)^{1/2}.$$

The chosen subsequence consequently converges in operator norm to a self-adjoint B with $\|B\|_{\text{op}} \leq \kappa$. This subsequence, and hence B , may depend on the point x .

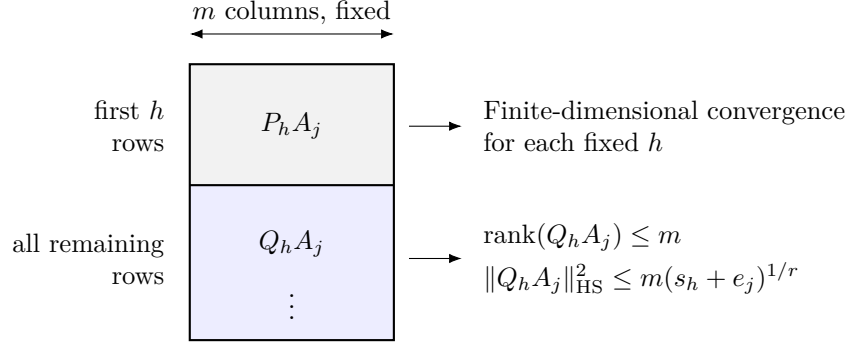


Figure 1: The number of rows can grow with the number of negative eigenfunctions, but the number of physical channels remains m . After fixing the first h rows, the identity $K_j^2 \approx B_j^2 + (A_j A_j^T)^r$ bounds the operator norm in the remaining rows. The rank bound converts it into a Hilbert–Schmidt bound, which tends to zero as first $j \rightarrow \infty$ and then $h \rightarrow \infty$. The drawing is schematic; all matrices are padded by zero rows in $\mathcal{H}$.

We can now pass to all the required products. Hilbert–Schmidt convergence of A_j implies operator norm convergence of $A_j A_j^T$ and of $A_j^T A_j$. Continuous functional calculus for nonnegative operators then gives convergence of their r th powers. Together with (5.14) and (5.15), this proves

$$[B, \Lambda^2] = 0, \quad \Lambda^2 - B^2 = (A A^T)^r.$$

Also $B_j A_j \rightarrow B A$ in Hilbert–Schmidt norm. Taking its i th row and using both the derivative convergence and the first-order error in (5.13) gives $a'_i(x) = (B A)_{i,\cdot}$ for every i . Finally, continuity of the finite-dimensional Young deficit and $Y(W, A_j^T A_j) \rightarrow 0$ yield $Y(W, A^T A) = 0$. Lemma 2.2 gives $W = q(A^T A)^r$.

We have proved these statements at every point of one common set of full measure. The construction selects no function $x \mapsto B(x)$: only the existence of a suitable operator at each such point enters the next section. This completes the proof of Proposition 5.1.

6 Fixed channels and the equality profiles

The pointwise relations of Proposition 5.1 still involve an auxiliary operator B that may depend on the point without a prescribed measurable choice. We now extract constant subspaces from those relations. On each nonzero subspace, B becomes a uniquely determined matrix and satisfies an ordinary differential equation. This will identify every real equality potential; realification and a scalar calculation will then complete the proof of Theorem 1.1.

6.1 The channel subspaces are constant

Assume first that W is real and is a nonzero equality potential. Use the rows a_i , numbers k_i , and operator $\Lambda = \text{diag}(k_i)$ from Proposition 5.1. For each distinct value $\alpha > 0$ in the list (k_i) , let n_α be its finite multiplicity, and let G_α be the $n_\alpha \times m$ matrix whose rows are the a_i with $k_i = \alpha$. Each G_α has entries in $H^1(\mathbb{R})$, and we use their continuous representatives.

At every point in the common full-measure set of that proposition, choose one of the operators B it supplies. Since B commutes with Λ^2 , it preserves each of its eigenspaces. Moreover,

$$A A^T = (\Lambda^2 - B^2)^{1/r}$$

commutes with both B and Λ^2 . Writing B_α for the block on the α^2 -eigenspace therefore gives

$$G_\alpha G_\beta^T = 0 \quad (\alpha \neq \beta), \quad (6.1)$$

$$G'_\alpha = B_\alpha G_\alpha, \quad [B_\alpha, G_\alpha G_\alpha^T] = 0, \quad (6.2)$$

$$\alpha^2 I - B_\alpha^2 = (G_\alpha G_\alpha^T)^r, \quad \|B_\alpha\|_{\text{op}} \leq \kappa. \quad (6.3)$$

These statements hold almost everywhere and require no compatibility between the choices of B at different points.

Lemma 6.1. *For every α , the subspace*

$$\mathcal{R}_\alpha = (\ker G_\alpha(x))^\perp \subset \mathbb{R}^m$$

is independent of $x \in \mathbb{R}$. The nonzero spaces $\mathcal{R}_\alpha$ are mutually orthogonal, and

$$\sum_\alpha \dim \mathcal{R}_\alpha \leq m. \quad (6.4)$$

In particular, only finitely many G_α are nonzero. The potential W has a continuous representative, equal to zero on the orthogonal complement of these spaces and equal to $q(G_\alpha^T G_\alpha)^r$ on $\mathcal{R}_\alpha$. With this representative, every G_α is C^2 .

Proof. Fix $v \in \mathbb{R}^m$. Equation (6.2) implies

$$\|(G_\alpha v)'(x)\| \leq \kappa \|G_\alpha(x)v\| \quad \text{for almost every } x.$$

The function $G_\alpha v$ is locally absolutely continuous. If it vanishes at x_0 , integration and Gronwall's inequality, first to the right and then to the left of x_0 , show that it vanishes everywhere. Consequently $\ker G_\alpha(x)$ is independent of x , including points outside the original full-measure set. This proves the first assertion.

The products in (6.1) are continuous, so their almost-everywhere vanishing implies their vanishing everywhere. Thus the constant row spaces $\mathcal{R}_\alpha$ are mutually orthogonal. Any finite collection has total dimension at most m , proving (6.4) and finiteness of the nonzero blocks.

On the full-measure set from Proposition 5.1, $A^T A = \sum_\alpha G_\alpha^T G_\alpha$. Each summand acts on its own orthogonal space $\mathcal{R}_\alpha$, and only finitely many summands are nonzero. Equation (5.4) therefore yields

$$W = q \sum_\alpha (G_\alpha^T G_\alpha)^r \quad \text{almost everywhere.} \quad (6.5)$$

The right-hand side is continuous and has the asserted restrictions; we henceforth use it as the representative of W . The row eigenfunction equation (5.2) now reads

$$G''_\alpha = \alpha^2 G_\alpha - G_\alpha W \quad \text{in distributions.} \quad (6.6)$$

Its right-hand side is continuous. Integrating this identity twice shows that the continuous representative of G_α is C^2 . $\square$

The row spaces already split the physical space $\mathbb{R}^m$ into finitely many constant channels. To determine the potential within one of them, we also need a constant range for G_α in the eigenfunction index space $\mathbb{R}^{n_\alpha}$.

Lemma 6.2. *For each α with $\ell := \dim \mathcal{R}_\alpha > 0$, the range of $G_\alpha(x)$ in $\mathbb{R}^{n_\alpha}$ is independent of x . In constant orthonormal bases of this range and of $\mathcal{R}_\alpha$, the restriction of G_α is an invertible C^2 matrix $D : \mathbb{R} \rightarrow \mathbb{R}^{\ell \times \ell}$. The matrix $B_0 = D'D^{-1}$ is C^1 and satisfies, everywhere on $\mathbb{R}$,*

$$B_0 = B_0^T, \quad \alpha^2 I - B_0^2 = (DD^T)^r > 0, \quad B'_0 = r(B_0^2 - \alpha^2 I). \quad (6.7)$$

Proof. Choose a constant orthonormal basis of $\mathcal{R}_\alpha$, and let $E(x)$ be the matrix of the restriction of $G_\alpha(x)$ to this space. It has full column rank ℓ at every point, since the kernel of G_α is constant. The projection onto its range is

$$P(x) = E(x)(E(x)^T E(x))^{-1} E(x)^T.$$

The commutator in (6.2) shows that B_α preserves the range of $G_\alpha G_\alpha^T$, which is precisely the range of E . Hence $E' = B_\alpha E$ has its columns in this range almost everywhere. Differentiating the projection gives

$$\begin{aligned} P' &= (I - P)E'(E^T E)^{-1}E^T \\ &\quad + E(E^T E)^{-1}(E')^T(I - P) = 0 \quad \text{almost everywhere.} \end{aligned}$$

Since P is C^1 , it is constant.

Choose a constant orthonormal basis of this range as well. The resulting square matrix D is invertible everywhere and C^2 . Almost everywhere, $D'D^{-1}$ is the restriction of B_α to its invariant range in these coordinates. Equations (6.2) and (6.3) give the symmetry and square identity in (6.7) almost everywhere; continuity extends them to every point. Positivity is strict because D is invertible.

On $\mathcal{R}_\alpha$, Equation (6.5) identifies the potential with $q(D^T D)^r$. Equation (6.6) consequently becomes

$$D'' = \alpha^2 D - qD(D^T D)^r = \alpha^2 D - q(DD^T)^r D.$$

The second equality follows, for example, by a singular-value decomposition. Differentiating $B_0 = D'D^{-1}$ and using $q = r + 1$ now gives

$$B'_0 = D''D^{-1} - B_0^2 = \alpha^2 I - q(\alpha^2 I - B_0^2) - B_0^2 = r(B_0^2 - \alpha^2 I),$$

as claimed. $\square$

6.2 The matrix Riccati equation separates the profiles

Fix a nonzero block and the matrices of Lemma 6.2. Diagonalize the symmetric matrix $B_0(x_*)$ at one point by a constant orthogonal change of range coordinates. The right-hand side of its Riccati equation is a polynomial in B_0 . Uniqueness for this matrix ordinary differential equation therefore preserves the diagonal form. More explicitly, the strict inequality in (6.7) places each initial diagonal entry in $(-\alpha, \alpha)$. There is a unique $y_j \in \mathbb{R}$ such that the scalar solution with that initial value is

$$b_j(x) = -\alpha \tanh(r\alpha(x - y_j)), \quad 1 \leq j \leq \ell. \quad (6.8)$$

The diagonal matrix with these entries solves the same initial-value problem as B_0 on all of $\mathbb{R}$, so uniqueness gives equality everywhere.

Equation (6.7) implies that DD^T is diagonal in these coordinates. Since $D' = \text{diag}(b_j)D$, each row of D is a positive scalar multiple of its value at x_* . None of these rows is zero, and their

directions are mutually orthogonal because DD^T is diagonal. Taking their normalized directions as a constant orthonormal basis of $\mathcal{R}_\alpha$ makes D diagonal with positive entries. The potential in this fixed basis is therefore diagonal, with entries

$$q(\alpha^2 - b_j(x)^2) = (r+1)\alpha^2 \operatorname{sech}^2(r\alpha(x - y_j)). \quad (6.9)$$

Combining these bases over the finitely many nonzero spaces $\mathcal{R}_\alpha$, and adding a basis of their orthogonal complement, proves the required necessity for real W . There are at most m nonzero profiles by (6.4). The zero potential has the same description with no nonzero profile.

6.3 Complex Hermitian potentials

For a complex Hermitian potential W , take the realification $\widetilde{W}$ from (4.2). The proof of Proposition 4.1 shows that it is equivalent to $W \oplus \overline{W}$ by a constant unitary matrix and that both the trace-power integral and the negative spectral moment are doubled. Thus equality for W implies equality for the real potential $\widetilde{W}$.

The real result provides a constant unitary matrix V such that

$$W(x) \oplus \overline{W}(x) = V \operatorname{diag}(f_1(x), \dots, f_{2m}(x)) V^* \quad \text{almost everywhere,}$$

where each f_j is either a profile of the form (6.9) or zero. Let P be the projection onto the first summand $\mathbb{C}^m$ and put $\widehat{P} = V^* P V$. Since P commutes with $W \oplus \overline{W}$,

$$(f_i(x) - f_j(x)) \widehat{P}_{ij} = 0 \quad \text{almost everywhere.}$$

Consequently $\widehat{P}_{ij} = 0$ whenever the functions f_i, f_j are not equal almost everywhere. Partition the indices into groups of identical functions, including the possible zero function. The range of $\widehat{P}$ splits as the orthogonal sum of its intersections with these coordinate groups. On each group the diagonal potential is a scalar multiple of the identity, so any constant orthonormal basis of that intersection diagonalizes its restriction. Transporting these bases by V to the first summand gives a constant unitary basis of $\mathbb{C}^m$ in which W has the asserted profiles. This proves necessity for complex Hermitian potentials.

6.4 Each listed potential attains equality

It remains to verify attainment, including the exact spectral moment, without appealing to a classification of scalar optimizers. Fix $a > 0$ and $x_0 \in \mathbb{R}$, and set

$$V(x) = (r+1)a^2 \operatorname{sech}^2(ra(x - x_0)), \quad H = -\partial_x^2 - V.$$

The first-order factorization used in the commutation argument of [2, Section II] also explains why this profile has only one negative eigenvalue. Put

$$w(x) = a \tanh(ra(x - x_0)), \quad Q = \partial_x + w, \quad \operatorname{Dom} Q = H^1(\mathbb{R}).$$

Then $Q^* = -\partial_x + w$ on $H^1(\mathbb{R})$. Since w and w' are bounded, both product domains are $H^2(\mathbb{R})$, and direct differentiation yields

$$Q^* Q = H + a^2, \quad (6.10)$$

$$Q Q^* = -\partial_x^2 + a^2 + (r-1)a^2 \operatorname{sech}^2(ra(x - x_0)) \geq a^2. \quad (6.11)$$

Here the last inequality uses $r > 1$. The equation $Q\psi = 0$ has the one-dimensional L^2 solution space

$$\ker Q = \text{span} \left\{ \text{sech}^{1/r}(ra(x - x_0)) \right\}.$$

Thus $-a^2$ is a simple eigenvalue of H , and (6.10) excludes eigenvalues below it. If $H\phi = \lambda\phi$ with $-a^2 < \lambda < 0$, put $\mu = \lambda + a^2$ and $g = Q\phi$. Then $\|g\|_2^2 = \mu\|\phi\|_2^2 > 0$. Moreover, $Q^*Q\phi = \mu\phi \in \text{Dom } Q$, so $g \in \text{Dom}(QQ^*)$ and $QQ^*g = \mu g$. This contradicts (6.11), since $\mu < a^2$. By Lemma 2.1, all negative spectrum is discrete. Hence $-a^2$ is the sole negative eigenvalue.

Finally, $2p - 1 = 2\gamma$ and the substitution $s = \tanh y$ give

$$\begin{aligned} \int_{\mathbb{R}} V(x)^p dx &= \frac{(r+1)^p}{r} a^{2\gamma} \int_{\mathbb{R}} \text{sech}^{2p} y dy \\ &= \frac{(r+1)^p}{r} a^{2\gamma} \int_{-1}^1 (1-s^2)^{p-1} ds = \frac{b_\sigma}{D_\sigma} a^{2\gamma}, \end{aligned}$$

where $(r+1)^p/r = D_\sigma^{-1}$ and $p-1 = \sigma$. Consequently

$$\text{Tr}(H)_-^\gamma = a^{2\gamma} = C_\gamma \int_{\mathbb{R}} V^p.$$

The zero potential contributes zero to both sides. Finite orthogonal direct sums add the spectral moments and trace-power integrals, and a constant unitary conjugation leaves both unchanged. Every potential listed in Theorem 1.1 therefore attains equality, completing its proof.

A The exact integrated trace deficit

The matrix comparison needed in the action argument has an exact remainder. We give the proof from [11, Section 3], retaining both nonnegative terms. Together they control the commutator and the algebraic equality relation used in the action argument. Throughout this appendix, $0 < \sigma < 1$, $\beta = \sigma - \frac{1}{2}$, and $r = 1/\sigma$, $p = 1 + \sigma$, $q = 1 + r$. Define $c_\sigma > 0$ by

$$c_\sigma^{-1} = \int_0^\infty t^\beta (t^{-1/2} - (t+1)^{-1/2}) dt. \quad (\text{A.1})$$

The integrand is $O(t^{\sigma-1})$ at zero and $O(t^{\sigma-2})$ at infinity. Scaling therefore gives

$$c_\sigma \int_0^\infty t^\beta (t^{-1/2} - (t+y)^{-1/2}) dt = y^\sigma, \quad y \geq 0. \quad (\text{A.2})$$

Theorem A.1 (Exact trace deficit). *Let $n \geq 1$ and let $B, V \in \mathbb{C}^{n \times n}$ be Hermitian. Suppose that*

$$Y_s = (sI - B)^2, \quad X_s = Y_s + V \geq bI \quad (s \in \mathbb{R}) \quad (\text{A.3})$$

for some $b > 0$. Set

$$R(t) = \frac{1}{\pi} \int_{\mathbb{R}} (X_s + tI)^{-1} ds, \quad P(t) = t^{-1/2}I - R(t), \quad t > 0, \quad (\text{A.4})$$

and

$$N = c_\sigma \int_0^\infty t^\beta P(t) dt. \quad (\text{A.5})$$

These integrals converge absolutely in matrix norm, and $R(t) > 0$. There are continuous Hermitian matrices $L(t) = R(t)^{-1} > 0$ and $Z(t)$ satisfying

$$V + tI = L^2 + Z^2 + i[B, Z], \quad LZ + ZL + i[B, L] = 0. \quad (\text{A.6})$$

If $N \geq 0$, then

$$\text{tr}(NV) - \text{tr}(N^q) = c_\sigma \int_0^\infty t^\beta \left(\text{tr}(RZ^2) + q \|R^{-1/2} - R^{1/2}(tI + N^r)^{1/2}\|_{\text{HS}}^2 \right) dt. \quad (\text{A.7})$$

In particular, $\text{tr}(NV) \geq \text{tr}(N^q)$.

The hypothesis concerns the full symbol X_s ; the perturbation V itself need not be positive. The proof first recovers L and Z from solutions on the two half-lines, then evaluates a logarithmic integral, and finally completes a matrix square.

A.1 Half-line boundary maps

Quadratic growth in s , together with (A.3), gives an integrable bound for X_s^{-1} . Hence $R(t) > 0$ and $R(t) = O(1)$ as $t \downarrow 0$. Diagonalization of B gives

$$\frac{1}{\pi} \int_{\mathbb{R}} (Y_s + tI)^{-1} ds = t^{-1/2} I.$$

For $t \geq 1$, both $X_s + tI$ and $Y_s + tI$ dominate $c(s^2 + t)I$, with $c > 0$ independent of s, t . The resolvent identity bounds their inverse difference by $C(s^2 + t)^{-2}$, and therefore

$$\|P(t)\|_{\text{op}} = \begin{cases} O(t^{-1/2}), & t \downarrow 0, \\ O(t^{-3/2}), & t \rightarrow \infty. \end{cases} \quad (\text{A.8})$$

These bounds prove the asserted convergence of N .

Fix $t > 0$ and put $\nabla = \partial_x - iB$. We shall use the equation

$$(-\nabla^2 + V + tI)\psi = 0. \quad (\text{A.9})$$

Matrix Riccati equations also appear in the commutation method for Schrödinger inequalities [2]. Here their role is to compute the integrated resolvent through boundary data.

For each $v \in \mathbb{C}^n$ there is a unique H^1 solution of (A.9) on either half-line with $\psi(0) = v$. Indeed, the sesquilinear form

$$a_t(\phi, \psi) = \int (\langle \nabla \phi, \nabla \psi \rangle + \langle \phi, (V + tI)\psi \rangle) dx,$$

with inner products linear in the second variable, is coercive on $H^1(\mathbb{R}; \mathbb{C}^n)$: its Fourier symbol $X_s + tI$ dominates $c_t(1 + s^2)I$. Zero extension gives coercivity on the trace-zero subspace of either half-line. Correcting any H^1 extension of v by the Riesz representation theorem gives the unique weak solution. The equation implies $\psi'' \in L^2$, so this solution lies in H^2 ; both ψ and ψ' vanish at the infinite end.

Define the right and left boundary matrices S, T by

$$\nabla \psi(0+) = -Sv, \quad \nabla \psi(0-) = Tv. \quad (\text{A.10})$$

Integration by parts gives $a_t(\psi_v, \psi_w) = \langle v, Sw \rangle$ on the right and $a_t(\psi_v, \psi_w) = \langle v, Tw \rangle$ on the left. Thus S, T are Hermitian. Translation and uniqueness show that $\nabla\psi = -S\psi$ throughout the right half-line and $\nabla\psi = T\psi$ throughout the left half-line. Substituting these relations into (A.9), and allowing arbitrary boundary data, gives

$$V + tI = S^2 + i[B, S] = T^2 - i[B, T]. \quad (\text{A.11})$$

Glue the two solutions with boundary value v at zero. The resulting continuous function has derivative jump $-(S + T)v$, and hence satisfies

$$(-\nabla^2 + V + tI)\psi = \delta_0(S + T)v.$$

Fourier inversion, with forward kernel e^{-isx} , gives

$$v = \frac{1}{2\pi} \int_{\mathbb{R}} (X_s + tI)^{-1} ds (S + T)v = \frac{1}{2} R(t)(S + T)v.$$

The inverse symbol is $O(s^{-2})$, so its Fourier inverse is continuous and this evaluation at zero is justified. Therefore

$$L := \frac{1}{2}(S + T) = R^{-1} > 0, \quad Z := \frac{1}{2}(S - T)$$

satisfy (A.6), by adding and subtracting (A.11). The second equation in (A.6) determines Z uniquely from L , since $L > 0$; in an eigenbasis of L , the map $Z \mapsto LZ + ZL$ multiplies its (j, k) entry by the positive number $L_{jj} + L_{kk}$. This also proves the continuity of $Z(t)$, because $R(t)$, and hence $L(t)$, is continuous.

The diagonal entries of that same equation give $Z_{jj} = 0$ in an eigenbasis of L , so $\text{tr } Z = 0$. Multiplication on both sides by $R = L^{-1}$ gives $i[B, R] = -(ZR + RZ)$. Cyclicity of the trace now yields

$$\text{tr}(R(V + tI)) = \text{tr } L - \text{tr}(RZ^2). \quad (\text{A.12})$$

In particular, the correction $\text{tr}(RZ^2) = \text{tr}(ZRZ)$ is nonnegative. We will also need the location of the spectrum of $B + iS$. The right solutions obey $\psi' = i(B + iS)\psi$. For an eigenvector with eigenvalue z , the solution is $e^{izx}v$; its square integrability forces $\text{Im } z > 0$.

A.2 A logarithmic integral

The next step converts the boundary matrices into a scalar identity that can be integrated in t . Define

$$F(t) = \frac{\text{tr } V}{\sqrt{t}} - \frac{1}{\pi} \int_{\mathbb{R}} \log \frac{\det(X_s + tI)}{\det(Y_s + tI)} ds. \quad (\text{A.13})$$

The determinants are positive, and the logarithmic ratio is $O(|s|^{-2})$ at infinity. Its derivative in t is the trace of the resolvent difference, with an integrable bound locally uniform for $t > 0$. Thus the integral converges absolutely and F is continuously differentiable.

The first identity in (A.11) factors the symbol as

$$X_s + tI = (sI - B + iS)(sI - B - iS).$$

If $u_j + iv_j$ are the eigenvalues of $B + iS$, counted with algebraic multiplicity, then $v_j > 0$ and

$$\det(X_s + tI) = \prod_{j=1}^n ((s - u_j)^2 + v_j^2).$$

The denominator in (A.13) has the same form, with centers the eigenvalues of B and widths $\sqrt{t}$. Translations of the individual logarithms change their symmetric-cutoff integrals by $o(1)$: for $h(s) = \log(s^2 + v^2)$,

$$\frac{d}{da} \int_{-M}^M h(s-a) ds = h(M+a) - h(M-a) \longrightarrow 0$$

uniformly for bounded a . After centering the factors, the identity

$$\int_{\mathbb{R}} \log \frac{s^2 + v^2}{s^2 + w^2} ds = 2\pi(v - w), \quad v, w > 0,$$

follows by differentiation in v and integration from w to v . Since $\sum_j v_j = \text{tr } S = \text{tr } L$, we obtain

$$F(t) = \frac{\text{tr } V}{\sqrt{t}} - 2 \text{tr}(L - \sqrt{t}I). \quad (\text{A.14})$$

Set

$$G(t) = \text{tr}(L - 2\sqrt{t}I + tR). \quad (\text{A.15})$$

Both functions are nonnegative. For G , apply $\lambda - 2\sqrt{t} + t/\lambda \geq 0$ to the positive eigenvalues λ of L . For F , take the trace of (A.6) and use (A.14) to obtain

$$F(t) = t^{-1/2} \text{tr}((L - \sqrt{t}I)^2 + Z^2) \geq 0.$$

Equations (A.12) and (A.14) give the identity

$$\text{tr}(PV) = F + G + \text{tr}(RZ^2). \quad (\text{A.16})$$

Every term on the right is nonnegative. The bounds (A.8) therefore show that F , G , and $\text{tr}(RZ^2)$ are integrable against $t^\beta dt$, and that $t^{\beta+1}F(t)$ vanishes at both endpoints.

Differentiating (A.13) and using (A.14)–(A.15) gives

$$F'(t) = -\frac{\text{tr } V}{2t^{3/2}} + \text{tr } P(t), \quad tF'(t) = -\frac{1}{2}F(t) - G(t).$$

Integration by parts, with the endpoint limits just proved, yields $\int_0^\infty t^\beta G dt = \sigma \int_0^\infty t^\beta F dt$. Consequently, integration of (A.16) gives

$$\text{tr}(NV) = qc_\sigma \int_0^\infty t^\beta G(t) dt + c_\sigma \int_0^\infty t^\beta \text{tr}(RZ^2) dt. \quad (\text{A.17})$$

It remains to express the first integral as $\text{tr}(N^q)/q$ plus a nonnegative remainder.

A.3 The noncommutative square

For a positive semidefinite matrix D , let

$$F_0(t, D) = \frac{\text{tr } D}{\sqrt{t}} - 2 \text{tr}((tI + D)^{1/2} - \sqrt{t}I).$$

For a scalar eigenvalue $d \geq 0$, the corresponding summand is $\int_0^d (t^{-1/2} - (t+y)^{-1/2}) dy$. Tonelli's theorem and (A.2) therefore imply

$$c_\sigma \int_0^\infty t^\beta F_0(t, D) dt = \frac{1}{p} \text{tr}(D^p). \quad (\text{A.18})$$

Now expand the Hilbert–Schmidt square

$$\begin{aligned}\mathcal{D}_D(t) &:= \|R^{-1/2} - R^{1/2}(tI + D)^{1/2}\|_{\text{HS}}^2 \\ &= \text{tr}(R^{-1} + R(tI + D) - 2(tI + D)^{1/2}) \\ &= G(t) - \text{tr}(P(t)D) + F_0(t, D).\end{aligned}\tag{A.19}$$

No commutation is used: the cross terms cancel adjacent factors $R^{-1/2}R^{1/2}$, and cyclicity changes the remaining quadratic term into $\text{tr}(R(tI + D))$. All three terms in the last line are integrable by (A.8), (A.16), and (A.18). Thus

$$c_\sigma \int_0^\infty t^\beta G(t) dt = \text{tr}(ND) - \frac{1}{p} \text{tr}(D^p) + c_\sigma \int_0^\infty t^\beta \mathcal{D}_D(t) dt.\tag{A.20}$$

When $N \geq 0$, choose $D = N^r$. Since $rp = q = 1 + r$, the first two terms on the right equal $\text{tr}(N^q)/q$. Substitution into (A.17) proves (A.7) and completes the proof of Theorem A.1.

If B and V commute, the symbol hypothesis implies $V \geq bI$. Simultaneous diagonalization then gives $R(t) = (tI + V)^{-1/2}$, $N = V^\sigma$, and $Z = 0$. Both remainders vanish, as required by scalar equality.

B The regularized field and connecting paths

The finite-interval action argument requires a symmetric matrix path from K to $-K$ whose derivative measures the difference between a matrix field and $AA^\top$. We construct both the field and the path here. These are the constructions of [11, Sections 4–5]; we include their proofs to make the analytic input to the equality argument self-contained. The positivity statement below also identifies the resolvent to which Theorem A.1 applies.

Throughout this appendix, $K = \text{diag}(k_1, \dots, k_n)$ with $k_i > 0$, $\kappa = \max_i k_i$, and Sym_n denotes the real symmetric matrices. We use $\beta = \sigma - \frac{1}{2}$ and the constant c_σ from (A.1).

B.1 A field with a scalar boundary integral

For $\delta > 0$, define on $y \geq 0$

$$f_\delta(y) = \frac{c_\sigma}{\pi} \int_0^\infty t^\beta \left(\frac{1}{t + \delta} - \frac{1}{t + \delta + y} \right) dt,\tag{B.1}$$

and extend to $y < 0$ by the tangent line $f'_\delta(0)y$. The resulting function is C^1 , strictly increasing, and concave. Indeed, for $y \geq 0$,

$$f'_\delta(y) = a_\sigma(y + \delta)^{\beta-1}, \quad a_\sigma = \frac{c_\sigma}{\pi} \int_0^\infty \frac{u^\beta}{(1+u)^2} du > 0.\tag{B.2}$$

Both integrals converge since $-\frac{1}{2} < \beta < \frac{1}{2}$. The extension allows the functional calculus below even when its matrix argument has negative eigenvalues.

Set

$$\begin{aligned}Q_s(B) &= s^2 I - 2sB + K^2, \\ M_\delta(B) &= \lim_{R \rightarrow \infty} \int_{-R}^R (f_\delta(Q_s(B)) - f_\delta(s^2)I) ds, \quad B \in \text{Sym}_n,\end{aligned}\tag{B.3}$$

where the symmetric cutoff cancels the leading term $-2sf'_\delta(s^2)B$, which is odd in s . The scalar counterpart is

$$\mu_\delta(z) = \int_{\mathbb{R}} (f_\delta(s^2 + z) - f_\delta(s^2)) ds, \quad z \in \mathbb{R}. \quad (\text{B.4})$$

Proposition B.1 (Regularized field). *The limit in (B.3) exists locally uniformly, and $M_\delta : \text{Sym}_n \rightarrow \text{Sym}_n$ is locally Lipschitz. There is a C^1 function $J_\delta : \text{Sym}_n \rightarrow \mathbb{R}$, normalized by $J_\delta(0) = 0$, such that*

$$dJ_\delta(B)[H] = -\text{tr}(M_\delta(B)H). \quad (\text{B.5})$$

The scalar function μ_δ is locally Lipschitz and strictly increasing, with $\mu_\delta(0) = 0$, and

$$\mu_\delta(z) = (z + \delta)^\sigma - \delta^\sigma \quad (z \geq 0). \quad (\text{B.6})$$

For every real unit vector e , and for every standard basis vector e_i , respectively,

$$e^\top M_\delta(B)e \leq \mu_\delta(e^\top K^2 e - (e^\top B e)^2), \quad (\text{B.7})$$

$$B e_i = y e_i \implies M_\delta(B) e_i = \mu_\delta(k_i^2 - y^2) e_i. \quad (\text{B.8})$$

For each diagonal sign matrix S ,

$$M_\delta(SBS) = SM_\delta(B)S. \quad (\text{B.9})$$

Finally,

$$\begin{aligned} J_\delta(-K) - J_\delta(K) &= \sum_{i=1}^n \int_{-k_i}^{k_i} \mu_\delta(k_i^2 - s^2) ds \\ &\longrightarrow b_\sigma \sum_{i=1}^n k_i^{2\sigma+1} \quad (\delta \downarrow 0). \end{aligned} \quad (\text{B.10})$$

Proof. We first justify the matrix integral, including its local Lipschitz dependence. If a scalar function h is Lipschitz on an interval containing the spectra of Hermitian matrices X, Y , then

$$\|h(X) - h(Y)\|_{\text{HS}} \leq \text{Lip}(h) \|X - Y\|_{\text{HS}}. \quad (\text{B.11})$$

To see this, use eigenbases e_i of X and v_j of Y . The mixed matrix entries of the two differences are $(h(\lambda_i) - h(\nu_j))\langle e_i, v_j \rangle$ and $(\lambda_i - \nu_j)\langle e_i, v_j \rangle$; square the scalar Lipschitz inequality and sum over i, j .

On a bounded set of B 's the spectrum of $E_s(B) = -2sB + K^2$ is contained in $[-a|s|, a|s|]$ for all sufficiently large $|s|$. For

$$h_s(y) = f_\delta(s^2 + y) - f_\delta(s^2) - f'_\delta(s^2)y,$$

Equation (B.2) gives $\sup_{|y| \leq a|s|} |h'_s(y)| = O(|s|^{2\beta-3})$. It follows from (B.11) that

$$f_\delta(Q_s(B)) - f_\delta(s^2)I = -2sf'_\delta(s^2)B + T_s(B),$$

where both $\|T_s(B)\|_{\text{HS}}$ and the local Lipschitz constant of T_s are $O(|s|^{2\beta-2})$. For the first bound, include $f'_\delta(s^2)K^2$ in T_s ; for the second, apply (B.11) to $h_s(E_s(B))$ and use $E_s(B_1) - E_s(B_2) = -2s(B_1 - B_2)$. Because $2\beta - 2 < -1$, these remainders are integrable at infinity. The odd term cancels at each symmetric cutoff. On bounded s -intervals, (B.11) applies directly to f_δ . This proves the asserted convergence and regularity.

For the primitive, choose a scalar antiderivative $g'_\delta = f_\delta$. Trace differentiation gives $d \operatorname{tr}(g_\delta(X))[H] = \operatorname{tr}(f_\delta(X)H)$. This formula follows for polynomials by cyclicity and for g_δ by approximation in C^1 on compact spectral intervals. For $s \neq 0$, the function

$$B \mapsto \frac{\operatorname{tr}(g_\delta(Q_s(B)) - g_\delta(K^2))}{2s} + f_\delta(s^2) \operatorname{tr} B$$

has differential $-\operatorname{tr}((f_\delta(Q_s(B)) - f_\delta(s^2)I)H)$. At $s = 0$ that differential is constant in B and also has a primitive. Thus the negative of every finite-cutoff field has zero integral around each closed piecewise smooth curve in Sym_n . Local uniform convergence passes this property to $-M_\delta$. Path integration, starting at zero, defines J_δ and proves (B.5).

For bounded z , the integrand in (B.4) and its Lipschitz constant in z are $O(|s|^{2\beta-2})$ at infinity. This proves absolute convergence and local Lipschitz continuity of μ_δ . Strict increase and $\mu_\delta(0) = 0$ follow from the same properties of f_δ . For $z \geq 0$, Tonelli's theorem applied to (B.1) gives

$$\mu_\delta(z) = c_\sigma \int_0^\infty t^\beta ((t + \delta)^{-1/2} - (t + \delta + z)^{-1/2}) dt.$$

The normalization (A.1), after scaling, evaluates this as (B.6).

Translations in the variable s remain legitimate with a symmetric cutoff. More precisely, for fixed $a, z \in \mathbb{R}$,

$$\lim_{R \rightarrow \infty} \int_{-R}^R (f_\delta((s-a)^2 + z) - f_\delta(s^2)) ds = \mu_\delta(z). \quad (\text{B.12})$$

Indeed $F_z(s) = f_\delta(s^2 + z)$ is even and $F'_z(s) = O(|s|^{2\beta-1}) \rightarrow 0$ at infinity. Pairing the two short intervals produced by a translation bounds the difference between the shifted and unshifted cutoff integrals by

$$|a|^2 \sup_{R-|a| \leq u \leq R+|a|} |F'_z(u)|,$$

which tends to zero.

Scalar concavity in an eigenbasis of $Q_s(B)$ now gives

$$e^\top f_\delta(Q_s(B))e \leq f_\delta(e^\top Q_s(B)e).$$

The scalar argument on the right is $(s - e^\top B e)^2 + e^\top K^2 e - (e^\top B e)^2$. Integration and (B.12) prove (B.7). If $B e_i = y e_i$, functional calculus gives equality in the corresponding coordinate, proving (B.8). Since $SK^2S = K^2$, conjugation inside the defining integral proves (B.9).

On diagonal matrices, (B.5) and (B.8) read

$$dJ_\delta(\operatorname{diag}(b_1, \dots, b_n)) = - \sum_i \mu_\delta(k_i^2 - b_i^2) db_i.$$

Integrating from K to $-K$ gives the first equality in (B.10). For $z \geq 0$, $0 \leq \mu_\delta(z) \leq z^\sigma$ and $\mu_\delta(z) \rightarrow z^\sigma$. Dominated convergence and the substitution $s = k_i u$ give its stated limit. $\square$

The field is defined on all symmetric matrices, but the action argument compares it with a positive matrix $AA^\top$. At such values its directional bound forces positivity of the entire quadratic symbol $Q_s(B)$. This is the condition needed for the trace deficit formula.

Proposition B.2 (Positive values of the field). *If $M = M_\delta(B) \geq 0$, then*

$$Q_s(B) \geq 0 \quad (s \in \mathbb{R}), \quad 0 \leq M \leq \kappa^{2\sigma} I. \quad (\text{B.13})$$

Set $V = K^2 - B^2 + \delta I$ and use B, V in (A.4) and (A.5). Then the symbol $(sI - B)^2 + V = Q_s(B) + \delta I$ is bounded below by δI , and

$$N = M + \delta^\sigma I. \quad (\text{B.14})$$

In particular, the hypotheses of Theorem A.1 hold.

Proof. By (B.7), positivity of M and strict increase of μ_δ imply $e^\top K^2 e \geq (e^\top B e)^2$ for every real unit vector e . Completing the scalar square gives

$$e^\top Q_s(B) e = (s - e^\top B e)^2 + e^\top K^2 e - (e^\top B e)^2 \geq 0.$$

Thus $Q_s(B) \geq 0$, also as a Hermitian matrix on $\mathbb{C}^n$. Moreover, (B.7) gives $M \leq \mu_\delta(\kappa^2)I \leq \kappa^{2\sigma}I$, using $(a + b)^\sigma \leq a^\sigma + b^\sigma$.

Write $Y_s = (sI - B)^2$. Diagonalization of B and (B.12) allow replacement of the scalar subtraction $f_\delta(s^2)I$ in (B.3) by $f_\delta(Y_s)$. Since both $Q_s(B)$ and Y_s are positive, their difference under f_δ is, by (B.1),

$$\frac{c_\sigma}{\pi} \int_0^\infty t^\beta \left((Y_s + (t + \delta)I)^{-1} - (Q_s(B) + (t + \delta)I)^{-1} \right) dt.$$

For fixed B, δ , both matrices being inverted are bounded below by $c(1 + t + s^2)I$. Their difference is $K^2 - B^2$, so the resolvent identity bounds the norm of the integrand by $Ct^\beta(1 + t + s^2)^{-2}$. Integrating first in s yields $C't^\beta(1 + t)^{-3/2}$, which is integrable on $(0, \infty)$. Fubini's theorem is therefore applicable and gives

$$M = c_\sigma \int_0^\infty t^\beta ((t + \delta)^{-1/2}I - R(t)) dt.$$

Subtracting this expression from (A.5) and scaling (A.1) gives (B.14). $\square$

B.2 Choosing the lower triangular coefficient matrix

Let $\mathcal{L}_n$ be the real lower triangular $n \times n$ matrices. Their dimension equals that of Sym_n . This dimension match allows us to prescribe a symmetric terminal value by choosing $C \in \mathcal{L}_n$. At a simple initial problem there is one choice up to signs of the rows. Continuation preserves the parity of this number, and hence produces the required terminal value.

Proposition B.3 (Connecting path). *Let $\Omega = (x_-, x_+)$ be bounded and let $U \in H_0^1(\Omega; \mathbb{R}^{n \times d})$, where $n, d \geq 1$, satisfy*

$$\int_\Omega UU^\top dx = I_n.$$

For every $\delta, \varepsilon > 0$ there are $C_\varepsilon \in \mathcal{L}_n$ and $B_\varepsilon \in C^1(\overline{\Omega}; \text{Sym}_n)$ such that, with $A_\varepsilon = C_\varepsilon U$,

$$\varepsilon B'_\varepsilon = M_\delta(B_\varepsilon) - A_\varepsilon A_\varepsilon^\top, \quad B_\varepsilon(x_-) = K, \quad B_\varepsilon(x_+) = -K. \quad (\text{B.15})$$

They satisfy

$$\sup_{x \in \overline{\Omega}} \|B_\varepsilon(x)\|_{\text{op}} \leq \kappa, \quad \sup_{0 < \varepsilon \leq 1} \|C_\varepsilon\|_{\text{HS}} < \infty, \quad (\text{B.16})$$

where the second bound is for fixed δ, K, Ω .

Proof. Use the continuous representative of U . Choose a smooth, nonnegative, compactly supported cutoff χ on Sym_n , equal to one on $\{B : \|B\|_{\text{op}} \leq \kappa\}$ and invariant under conjugation by diagonal sign matrices. Averaging a cutoff over the finite sign group produces such a function. The truncated field $\widehat{M} = \chi M_\delta$ is bounded and globally Lipschitz.

Fix $\varepsilon > 0$. For $C \in \mathcal{L}_n$ and $\theta \in [0, 1]$, the equation

$$\varepsilon B' = \theta \widehat{M}(B) - (CU)(CU)^\top, \quad B(x_-) = K \quad (\text{B.17})$$

has a unique C^1 solution on $\overline{\Omega}$. Define the continuous endpoint map

$$\Phi(C, \theta) = B(x_+) + K \in \text{Sym}_n.$$

Let $\mathcal{G}$ be the group of diagonal sign matrices, acting by $C \mapsto SC$ on $\mathcal{L}_n$ and by conjugation on Sym_n . Equivariance of the field and uniqueness of the ODE give

$$\Phi(SC, \theta) = S\Phi(C, \theta)S. \quad (\text{B.18})$$

We record the three facts needed for continuation: compactness of the zero set, freeness of the sign action there, and one regular orbit of zeros at $\theta = 0$. At a zero, integration of (B.17) gives

$$CC^\top = 2\varepsilon K + \theta \int_\Omega \widehat{M}(B(x)) dx. \quad (\text{B.19})$$

Consequently all zeros lie in a bounded set, and their joint set in $\mathcal{L}_n \times [0, 1]$ is compact. For $0 < \varepsilon \leq 1$, this calculation gives the uniform bound

$$\|C\|_{\text{HS}}^2 \leq 2 \text{tr } K + n|\Omega| \sup_{B \in \text{Sym}_n} \|\widehat{M}(B)\|_{\text{op}}. \quad (\text{B.20})$$

No row of C can vanish at a zero. Otherwise the reflection changing only row i would fix C ; ODE uniqueness and (B.18) would force $Be_i = ye_i$ throughout the interval. Equation (B.8) would then give

$$\varepsilon y' = \theta \chi(B(x)) \mu_\delta(k_i^2 - y^2), \quad y(x_-) = k_i.$$

The coefficient $\chi(B(x))$ is continuous and the right side is locally Lipschitz in y . Uniqueness forces $y \equiv k_i$, contradicting $y(x_+) = -k_i$. Thus the action of $\mathcal{G}$ on the zero set is free.

At $\theta = 0$ one has

$$\Phi(C, 0) = 2K - \varepsilon^{-1}CC^\top. \quad (\text{B.21})$$

Its lower triangular zeros are exactly $C = \text{diag}(d_1, \dots, d_n)$ with $d_i = \pm\sqrt{2\varepsilon k_i}$. Indeed, the first row of $CC^\top = 2\varepsilon K$ fixes $|d_1|$ and forces all other entries of the first column to vanish, and induction does the same for subsequent columns. These 2^n zeros form one sign orbit. At each, the derivative in a lower triangular direction H has diagonal entries $-2\varepsilon^{-1}d_i H_{ii}$ and lower off-diagonal entries $-\varepsilon^{-1}d_j H_{ij}$ for $i > j$. It is therefore an isomorphism $\mathcal{L}_n \rightarrow \text{Sym}_n$.

We next justify the continuation using only smooth approximation and the regular-value argument for parity [10, Sections 2–4]. Suppose there were no zero at $\theta = 1$. Compactness and freeness give a bounded invariant open set $O \subset \mathcal{L}_n$ containing the projection of every zero, whose closure lies where all rows are nonzero. There is a positive lower bound for $\|\Phi\|_{\text{HS}}$ on

$$(\partial O \times [0, 1]) \cup (\overline{O} \times \{1\}). \quad (\text{B.22})$$

Reparametrize θ by a smooth map equal to zero near zero and equal to one at one. The resulting map F agrees with (B.21) on an initial collar and retains the gap on (B.22).

Extend F constantly beyond the two ends of the parameter interval, smooth by convolution, and splice to (B.21) in a smaller initial collar. Averaging the approximation Ψ_0 in the form

$$\Psi(C, \theta) = \frac{1}{|\mathcal{G}|} \sum_{S \in \mathcal{G}} S \Psi_0(SC, \theta) S$$

gives a smooth equivariant approximation Ψ that still equals the initial map on that collar. It can be made uniformly close enough on $\overline{O} \times [0, 1]$ to preserve the positive gap.

Freeness permits an equivariant perturbation with arbitrary target directions. At each $C_0 \in \overline{O}$, choose a ball whose distinct sign translates are disjoint, and a smooth bump b supported in this ball with $b(C_0) = 1$. For $H \in \text{Sym}_n$ the map

$$V_H(C) = \sum_{S \in \mathcal{G}} b(SC) S H S$$

is equivariant and satisfies $V_H(C_0) = H$. Choose a basis of target matrices and then a finite collection of such neighborhoods covering $\overline{O}$. This gives finitely many smooth equivariant maps $V_1, \dots, V_N$ whose values span Sym_n at every point of $\overline{O}$.

Let $h \geq 0$ be smooth, vanish near zero, and be positive outside the collar where Ψ is the initial map. Choose it so that every zero of h lies within that collar, and set

$$\Psi_\alpha(C, \theta) = \Psi(C, \theta) + h(\theta) \sum_{j=1}^N \alpha_j V_j(C).$$

For α in a sufficiently small open ball, the gap on (B.22) persists. At a zero of this family with $0 < \theta < 1$, its full differential in (C, θ, α) is onto: when $h(\theta) > 0$ the parameter directions span the target, and when $h(\theta) = 0$ the derivative in C is the isomorphism computed for (B.21). Its interior zero set is consequently a smooth manifold. Sard's theorem [13], applied to projection of this manifold onto the parameter ball, gives arbitrarily small α for which zero is a regular value of $(C, \theta) \mapsto \Psi_\alpha(C, \theta)$. For completeness, if the full derivative is (L_1, L_2) in the (C, θ) and α directions, regularity of that projection means that each w admits v with $L_1 v + L_2 w = 0$. Hence $\text{ran } L_2 \subset \text{ran } L_1$, and surjectivity of (L_1, L_2) implies surjectivity of L_1 .

For such an α , the zero set in $O \times [0, 1]$ is a compact smooth one-dimensional manifold with boundary. The gap excludes boundary zeros except at $\theta = 0$, where the initial collar gives exactly the 2^n regular zeros above. The free sign action has a quotient that is again a compact one-dimensional manifold with boundary: a neighborhood disjoint from its nontrivial translates supplies the local interval or half-interval chart. All 2^n boundary points form one orbit, so the quotient has exactly one boundary point. This is impossible, since a compact one-dimensional manifold is a finite union of circles and closed intervals. Therefore the original map $\Phi(\cdot, 1)$ has a zero.

It remains to remove the cutoff. For the path at this zero and any fixed real unit vector e , put $y = e^\top B e$. Whenever $|y| > \kappa$, Equation (B.7) yields

$$e^\top M_\delta(B) e \leq \mu_\delta (e^\top K^2 e - y^2) < 0.$$

Since $\chi \geq 0$ and $(CU)(CU)^\top \geq 0$, the ODE implies $y' \leq 0$ there. Both endpoint values lie in $[-\kappa, \kappa]$. An excursion above κ cannot start from κ while y is nonincreasing; an excursion below $-\kappa$ cannot return to $-\kappa$ while y is nonincreasing. The two endpoint conditions therefore rule out both excursions. This holds for every e , so $\|B(x)\|_{\text{op}} \leq \kappa$. Consequently $\chi(B) = 1$, giving (B.15), and (B.20) supplies the remaining assertion in (B.16). $\square$

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